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PRACTICAL
HANDBOOK

VERIQTA

— ENGINEERING EXCELLENCE —

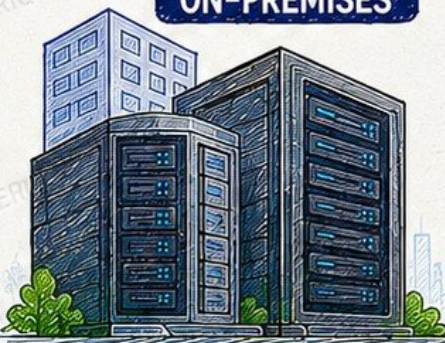


REAL WORLD
STRATEGIES
PRODUCTION
FOCUSED

AWS MIGRATION HANDBOOK

From On-Premises to Production on AWS

ON-PREMISES



AWS CLOUD



REDUCE RISK

Migrate with
confidence.



OPTIMIZE COSTS

Right-size and
modernize for
efficiency.



IMPROVE AGILITY

Accelerate innovation
and business
outcomes.



ENHANCE SECURITY

Build a secure
foundation in
the cloud.



**PROVEN
STRATEGIES**



**STEP-BY-STEP
GUIDE**



**REAL WORLD
EXAMPLES**



**PRODUCTION
CHECKLISTS**

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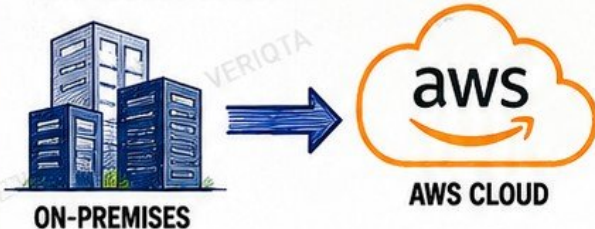


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1. INTRODUCTION TO AWS MIGRATION

WHAT CLOUD MIGRATION MEANS

Cloud migration is the process of moving applications, data, and business operations from on-premises infrastructure to the AWS Cloud to improve agility, scalability, resilience, and reduce costs.



MIGRATION BUSINESS DRIVERS



REDUCE COSTS

Lower infrastructure and operational expenses



IMPROVE AGILITY

Deploy faster and respond quickly to change



INCREASE RELIABILITY

Leverage AWS global infrastructure for high availability



SCALE ELASTICALLY

Scale resources up or down as needed



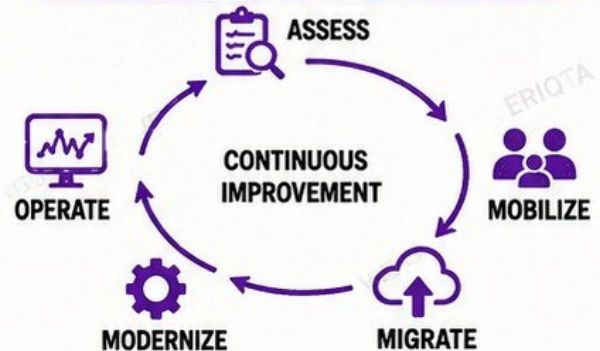
DRIVE INNOVATION

Use modern AWS services to build better solutions

COMMON MIGRATION CHALLENGES

- ⚠ LACK OF CLEAR STRATEGY AND PLANNING
- ⚠ UNDERSTANDING DEPENDENCIES AND COMPLEXITY
- ⚠ DATA VOLUME AND TRANSFER TIME
- ⚠ DOWNTIME AND BUSINESS CONTINUITY
- ⚠ SECURITY AND COMPLIANCE REQUIREMENTS
- ⚠ SKILL GAPS AND ORGANIZATIONAL READINESS
- ⚠ COST OVERRUNS AND UNEXPECTED EXPENSES

MIGRATION LIFECYCLE



ASSESS, MOBILIZE, MIGRATE AND MODERNIZE



ASSESS

Understand your current environment, dependencies, risks, and opportunities.



MOBILIZE

Build the foundation for migration: strategy, roadmap, team, and governance.



MIGRATE

Move workloads to AWS using the right migration strategies.



MODERNIZE

Optimize and modernize applications to improve performance, scalability, and business value.



KEY TAKEAWAY

AWS migration is a structured journey that starts with assessment, builds a strong foundation, moves workloads efficiently, and continuously modernizes for long-term success.



2. UNDERSTANDING THE EXISTING ENVIRONMENT



A DEEP UNDERSTANDING OF YOUR CURRENT ENVIRONMENT IS THE FOUNDATION OF A SUCCESSFUL AWS MIGRATION. INVENTORY, ANALYZE, AND PRIORITIZE BEFORE YOU MOVE.

APPLICATION INVENTORY



- LIST ALL APPLICATIONS
- CAPTURE PURPOSE AND FUNCTION
- IDENTIFY OWNERS AND STAKEHOLDERS
- RECORD TECHNOLOGY STACK
- NOTE INTEGRATIONS AND INTERFACES
- DOCUMENT LICENSE REQUIREMENTS

SERVER INVENTORY



- IDENTIFY ALL SERVERS
- RECORD OS AND VERSION
- CAPTURE CPU, MEMORY, STORAGE
- IDENTIFY SERVER ROLE
- NOTE UTILIZATION METRICS
- RECORD DEPENDENCIES

DATABASE INVENTORY



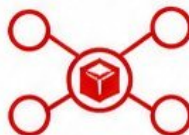
- LIST ALL DATABASES
- RECORD DATABASE ENGINE & VERSION
- CAPTURE SIZE AND GROWTH
- IDENTIFY DATABASE OWNER
- NOTE BACKUP AND RETENTION
- DOCUMENT DEPENDENCIES

NETWORK DEPENDENCIES



- IDENTIFY NETWORK TOPOLOGY
- DOCUMENT IP RANGES AND VLANs
- NOTE FIREWALL AND SECURITY RULES
- IDENTIFY VPN / DIRECT CONNECT
- RECORD DNS AND LOAD BALANCERS
- MAP INTERNET AND WAN LINKS

APPLICATION DEPENDENCIES



- MAP APPLICATION INTERACTIONS
- IDENTIFY UPSTREAM DEPENDENCIES
- IDENTIFY DOWNSTREAM DEPENDENCIES
- NOTE THIRD-PARTY SERVICES
- DOCUMENT DATA FLOWS
- IDENTIFY CRITICAL PATHS

BUSINESS CRITICALITY



- CLASSIFY BY BUSINESS IMPACT
- DEFINE RTO AND RPO REQUIREMENTS
- IDENTIFY PEAK USAGE PERIODS
- ASSESS TOLERANCE FOR DOWNTIME
- PRIORITIZE MIGRATION SEQUENCE
- ALIGN WITH BUSINESS OBJECTIVES



GOOD DISCOVERY = BETTER PLANNING = LOWER RISK = SUCCESSFUL MIGRATION
YOU CAN'T MIGRATE WHAT YOU DON'T UNDERSTAND.



VISIBILITY
KNOW YOUR
ENVIRONMENT



ACCURACY
BUILD A RELIABLE
INVENTORY



PRIORITIZATION
FOCUS ON WHAT
MATTERS MOST



CONFIDENCE
MAKE SMART
DECISIONS



SUCCESS
DELIVER BUSINESS
VALUE ON AWS

3. AWS MIGRATION ASSESSMENT



A COMPREHENSIVE ASSESSMENT HELPS YOU UNDERSTAND YOUR CURRENT STATE, IDENTIFY GAPS, AND CREATE A CLEAR, LOW-RISK MIGRATION STRATEGY.

1. MIGRATION READINESS



- ASSESS CLOUD ADOPTION MATURITY
- REVIEW CURRENT INFRASTRUCTURE
- CHECK APPLICATION COMPATIBILITY
- IDENTIFY SKILLS AND CAPABILITIES
- EVALUATE TOOLS AND PROCESSES
- DETERMINE OVERALL READINESS

2. TECHNICAL ASSESSMENT



- EVALUATE INFRASTRUCTURE
- ANALYZE APPLICATIONS AND DATA
- REVIEW DEPENDENCIES
- ASSESS SECURITY AND COMPLIANCE
- REVIEW NETWORK ARCHITECTURE
- IDENTIFY TECHNICAL GAPS

3. ORGANIZATIONAL READINESS



- ASSESS BUSINESS ALIGNMENT
- IDENTIFY STAKEHOLDERS
- EVALUATE CHANGE MANAGEMENT
- REVIEW GOVERNANCE STRUCTURE
- CHECK OPERATIONAL PROCESSES
- BUILD INTERNAL CLOUD CULTURE

4. TCO CONSIDERATIONS



- ESTIMATE CURRENT STATE COSTS
- FORECAST AWS CLOUD COSTS
- CONSIDER MIGRATION COSTS
- ANALYZE OPERATIONAL SAVINGS
- IDENTIFY COST OPTIMIZATION OPPORTUNITIES
- CALCULATE 3-YEAR TCO

5. MIGRATION RISKS



- IDENTIFY BUSINESS RISKS
- IDENTIFY TECHNICAL RISKS
- EVALUATE SECURITY RISKS
- ASSESS COMPLIANCE RISKS
- ANALYZE DATA AND INTEGRATION RISKS
- DEFINE RISK MITIGATION STRATEGIES

6. BUILDING THE MIGRATION BUSINESS CASE



- DEFINE BUSINESS OBJECTIVES
- QUANTIFY BENEFITS
- DOCUMENT COSTS AND SAVINGS
- EVALUATE ROI AND PAYBACK
- PREPARE EXECUTIVE SUMMARY
- GET STAKEHOLDER APPROVAL



STRONG ASSESSMENT = STRONG MIGRATION OUTCOME
ASSESS THOROUGHLY. PLAN SMARTER. MIGRATE WITH CONFIDENCE.



4. THE AWS 7 Rs MIGRATION STRATEGIES



THE 7 Rs HELP YOU CHOOSE THE RIGHT APPROACH FOR EACH WORKLOAD BASED ON YOUR BUSINESS GOALS, COST, COMPLEXITY, AND RISK.

1



REHOST
(LIFT AND SHIFT)

Move applications to AWS with minimal or no changes. Fastest approach with low risk and low cost.

2



REPLATFORM
(LIFT, TINKER AND SHIFT)

Make some optimizations to take advantage of AWS managed services without changing core architecture.

3



REFACTOR /
RE-ARCHITECT

Redesign or re-architect the application to improve scalability, performance, and cost efficiency.

4



REPURCHASE
(BUY)

Replace the application with a SaaS product. No need to manage infrastructure.

5



RELOCATE
(MOVE)

Move applications to AWS without changes and without taking advantage of AWS services. Rarely used, usually for licensing reasons.

6



RETAIN
(KEEP)

Keep the application in its current environment. Not every workload needs to move now.

7



RETIRE
(DECOMMISSION)

Decommission applications that are no longer needed. Eliminate cost and reduce risk.

CHOOSING THE CORRECT STRATEGY

KEY FACTORS TO CONSIDER

- ✓ BUSINESS OBJECTIVES
- ✓ APPLICATION COMPLEXITY
- ✓ COST AND ROI
- ✓ TIME TO MARKET
- ✓ RISK AND COMPLIANCE
- ✓ TEAM SKILLS AND CAPABILITIES



DECISION APPROACH

- 1 ASSESS EACH WORKLOAD
- 2 EVALUATE USING 7 Rs
- 3 COMPARE COST, RISK, AND BENEFIT
- 4 SELECT THE BEST STRATEGY
- 5 PLAN EXECUTION AND REVIEW



THE GOAL



SELECT THE RIGHT STRATEGY FOR THE RIGHT WORKLOAD TO MAXIMIZE VALUE ON AWS.

5. MIGRATION DISCOVERY AND DEPENDENCY MAPPING



**DISCOVER EVERYTHING. UNDERSTAND EVERYTHING.
MAP EVERYTHING. NOTHING IMPORTANT SHOULD BE MISSED
BEFORE YOU MIGRATE.**

1

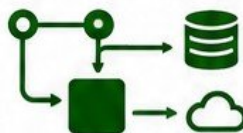
DISCOVERING SERVERS AND APPLICATIONS



- SCAN ON-PREMISES ENVIRONMENTS
- DISCOVER PHYSICAL, VIRTUAL, AND CLOUD RESOURCES
- IDENTIFY INSTALLED APPLICATIONS AND SERVICES
- CAPTURE SYSTEM CONFIGURATIONS
- COLLECT PERFORMANCE DATA

2

DEPENDENCY MAPPING



- MAP COMMUNICATION BETWEEN APPLICATIONS
- IDENTIFY DATABASE CONNECTIONS
- MAP NETWORK FLOW AND PORTS
- UNDERSTAND UPSTREAM AND DOWNSTREAM DEPENDENCIES
- VISUALIZE END-TO-END FLOW

3

APPLICATION GROUPING



- GROUP APPLICATIONS BY BUSINESS FUNCTION OR SERVICE
- CONSIDER DEPENDENCIES AND COMMUNICATION
- IDENTIFY MIGRATION CANDIDATES FOR EACH WAVE
- DEFINE MIGRATION SEQUENCING
- REDUCE RISKS AND DOWNTIME

4

IDENTIFYING HIDDEN DEPENDENCIES



- LOOK BEYOND DOCUMENTED DEPENDENCIES
- IDENTIFY HARD-CODED CONNECTIONS
- DETECT SHARED FILES AND SCRIPTS
- DISCOVER MANUAL PROCESSES
- VALIDATE WITH STAKEHOLDERS
- PREVENT SURPRISES DURING MIGRATION

5

AWS MIGRATION HUB



- CENTRALIZE DISCOVERY DATA
- AGGREGATE DATA FROM TOOLS AND AGENTS
- VISUALIZE APPLICATION DEPENDENCIES
- TRACK MIGRATION PROGRESS
- PRIORITIZE WORKLOADS
- COLLABORATE ACROSS TEAMS

6

DISCOVERY DATA



- INVENTORY REPORTS
- DEPENDENCY DIAGRAMS
- PERFORMANCE BASELINES
- COST AND UTILIZATION DATA
- BUSINESS CRITICALITY INPUT
- DATA DRIVES PLANNING AND DECISION MAKING



**COMPLETE AND ACCURATE DISCOVERY IS THE FOUNDATION
OF A SUCCESSFUL, LOW-RISK MIGRATION.**



BETTER VISIBILITY
KNOW WHAT YOU
HAVE AND HOW
IT CONNECTS



REDUCED RISK
IDENTIFY ISSUES
BEFORE THEY IMPACT
THE MIGRATION



SMARTER PLANNING
GROUP AND SEQUENCE
WORKLOADS
EFFECTIVELY



COST OPTIMIZATION
RIGHT-SIZE AND
PRIORITIZE FOR
MAXIMUM VALUE

6. DESIGNING THE AWS TARGET ARCHITECTURE



A WELL-DESIGNED TARGET ARCHITECTURE ENSURES SCALABILITY, SECURITY, HIGH AVAILABILITY, AND COST-EFFECTIVE OPERATIONS IN THE AWS CLOUD.

1



SOURCE VS TARGET ARCHITECTURE

- UNDERSTAND CURRENT (ON-PREMISES) ARCHITECTURE
- IDENTIFY GAPS AND IMPROVEMENT OPPORTUNITIES
- DESIGN FUTURE-STATE (AWS) ARCHITECTURE
- ALIGN WITH BUSINESS AND TECHNICAL GOALS

2



AWS REGIONS AND AVAILABILITY ZONES

- CHOOSE THE RIGHT AWS REGION
- LEVERAGE MULTIPLE AVAILABILITY ZONES (AZs)
- BUILD FOR RESILIENCY AND LOW LATENCY
- CONSIDER DATA RESIDENCY AND COMPLIANCE

3



VPC ARCHITECTURE

- DESIGN A SCALABLE AND SECURE VPC
- USE PUBLIC AND PRIVATE SUBNETS
- IMPLEMENT ROUTE TABLES AND GATEWAYS
- ENABLE NETWORK ISOLATION AND SEGMENTATION

4



COMPUTE SELECTION

- CHOOSE THE RIGHT COMPUTE SERVICES (EC2, ECS, EKS, LAMBDA, ETC.)
- MATCH WORKLOADS TO OPTIMAL SERVICES
- CONSIDER SCALABILITY AND COST
- DESIGN FOR PERFORMANCE AND RELIABILITY

5



STORAGE

- SELECT THE RIGHT STORAGE SERVICES (S3, EBS, EFS, FSx, GLACIER, ETC.)
- DESIGN FOR DURABILITY AND AVAILABILITY
- PLAN FOR BACKUP AND DATA LIFECYCLE
- OPTIMIZE COST AND PERFORMANCE

6



DATABASES

- CHOOSE BETWEEN MANAGED AND SELF-MANAGED (RDS, AURORA, DYNAMODB, ETC.)
- DESIGN FOR HIGH AVAILABILITY AND BACKUPS
- CONSIDER READ REPLICAS AND MULTI-AZ
- PLAN FOR SCALABILITY AND PERFORMANCE

7



HIGH AVAILABILITY

- DISTRIBUTE ACROSS MULTIPLE AZs
- USE ELASTIC LOAD BALANCING
- IMPLEMENT AUTO SCALING
- DESIGN FOR FAULT TOLERANCE AND RECOVERY

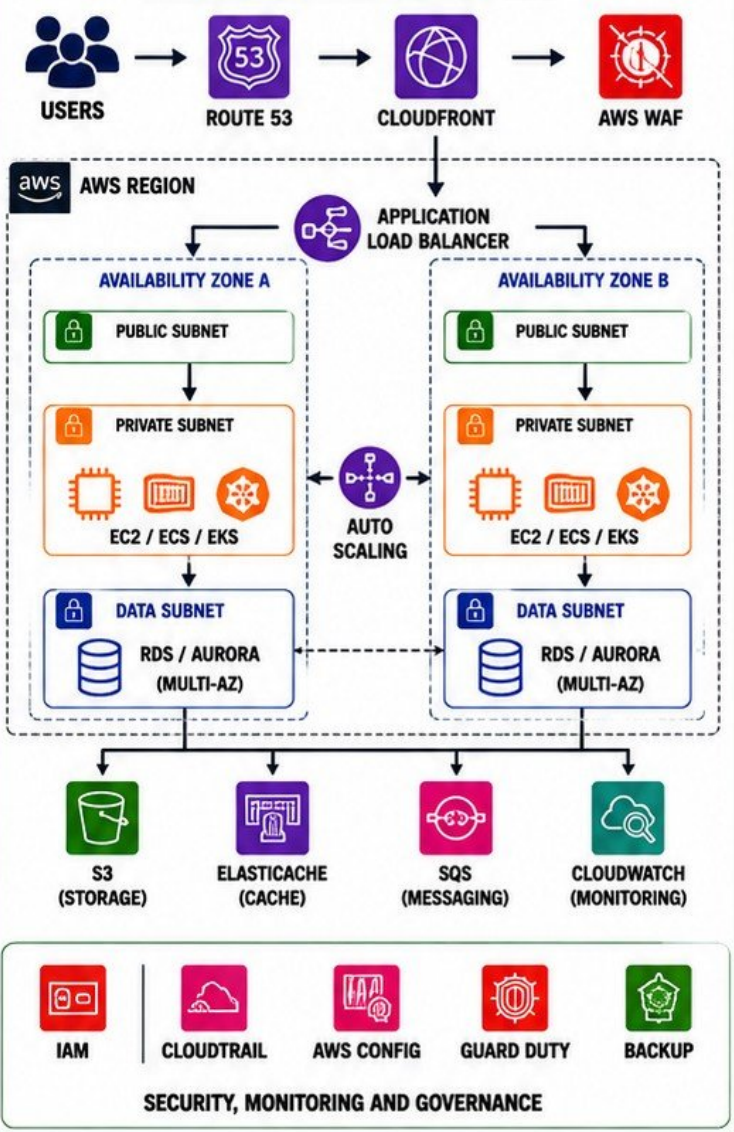
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TARGET ARCHITECTURE DIAGRAM

- VISUALIZE THE COMPLETE TARGET ARCHITECTURE
- SHOW COMPONENTS AND THEIR RELATIONSHIPS
- VALIDATE AGAINST BEST PRACTICES
- COMMUNICATE CLEARLY WITH STAKEHOLDERS

TARGET ARCHITECTURE DIAGRAM



A STRONG ARCHITECTURE TODAY BUILDS A SCALABLE, SECURE, AND RESILIENT FUTURE ON AWS.



7. BUILDING THE AWS LANDING ZONE



A WELL-ARCHITECTED LANDING ZONE PROVIDES A SECURE, GOVERNED, AND SCALABLE FOUNDATION FOR ALL WORKLOADS IN THE AWS CLOUD.

1 AWS ORGANIZATIONS



- Centrally manage multiple AWS accounts
- Enable consolidated billing and governance
- Apply policies across all accounts
- Build the foundation for scale

2 ACCOUNT STRUCTURE



- Define a clear account strategy
- Separate by environment, function, or workload
- Use OUs (Organizational Units) for grouping
- Follow multi-account best practices

3 AWS CONTROL TOWER



- Automate landing zone setup and governance
- Pre-configured best practice guardrails
- Account vending and lifecycle management
- Continuous compliance and updates

4 IDENTITY FOUNDATION



- Implement AWS IAM best practices
- Centralize identity with AWS IAM Identity Center (SSO)
- Enforce MFA and least privilege access
- Manage roles, groups, and permissions

5 CENTRALIZED LOGGING



- Centralize logs with Amazon CloudWatch
- Aggregate logs across accounts and Regions
- Enable log retention and encryption
- Integrate with Amazon S3 for long-term storage

6 SECURITY CONTROLS



- Enforce guardrails with AWS Organizations SCPs
- Enable AWS Config for compliance monitoring
- Use AWS Security Hub and Amazon GuardDuty
- Implement continuous security monitoring

7 NETWORKING FOUNDATION



- Design a secure, scalable network baseline
- Use AWS Transit Gateway for connectivity
- Implement hub-and-spoke VPC architecture
- Secure with AWS Network Firewall and VPC endpoints

8 GOVERNANCE



- Define policies and standards
- Tagging strategy
- Cost management guardrails
- Compliance frameworks
- Regular audits and continuous improvement



A STRONG LANDING ZONE ENSURES SECURITY, COMPLIANCE, OPERATIONAL EXCELLENCE, AND AGILITY FROM DAY ONE.



SECURE
BUILT-IN GUARDRAILS
AND BEST PRACTICES



SCALABLE
DESIGNED TO GROW
WITH YOUR BUSINESS



COMPLIANT
MEET REGULATORY
REQUIREMENTS



EFFICIENT
AUTOMATED AND
STANDARDIZED



COST OPTIMIZED
CONTROL COSTS
WITH GOVERNANCE

8. MIGRATION NETWORKING AND CONNECTIVITY



A ROBUST NETWORK FOUNDATION ENSURES SECURE, RELIABLE, AND HIGH-PERFORMANCE CONNECTIVITY BETWEEN YOUR ON-PREMISES ENVIRONMENT AND AWS.

1 VPC DESIGN



- Design a scalable and secure VPC
- Choose IP address ranges
- Use multiple Availability Zones
- Plan for future growth
- Enable DNS hostnames
- Enable DNS resolution

2 PUBLIC AND PRIVATE SUBNETS



- Use public subnets for internet-facing resources
- Use private subnets for application and data tiers
- Distribute subnets across multiple AZs
- Use NAT Gateway for outbound internet from private subnets

3 ROUTING



- Configure route tables
- Use internet gateway for public access
- Use NAT gateway for private subnet internet access
- Use transit gateway for advanced routing
- Implement route propagation

4 SITE-TO-SITE VPN



- Create a Virtual Private Gateway (VGW)
- Configure customer gateway on-premises
- Establish IPsec VPN tunnel
- Use static or BGP routing
- Test and validate VPN connectivity

5 AWS DIRECT CONNECT



- Establish private connection to AWS
- Use Direct Connect Location
- Configure Virtual Interface
- Use BGP for dynamic routing
- Lower latency and higher bandwidth

6 DNS CONSIDERATIONS



- Plan DNS architecture
- Use Amazon Route 53 for DNS
- Configure private hosted zones
- Implement DNS resolution between on-premises and AWS
- Plan for failover and latency

7 HYBRID CONNECTIVITY



- Combine VPN and Direct Connect for resiliency
- Use Transit Gateway for hub-and-spoke model
- Segment network for security
- Ensure seamless connectivity for workloads and users

8 NETWORK VALIDATION



- Validate end-to-end connectivity
- Test failover scenarios
- Monitor network performance
- Use Flow Logs and CloudWatch metrics
- Document and review regularly

TYPICAL HYBRID CONNECTIVITY ARCHITECTURE



STRONG NETWORK CONNECTIVITY IS THE BACKBONE
OF A SUCCESSFUL AND RELIABLE MIGRATION.



9. SECURITY, IDENTITY,
AND COMPLIANCE

**BUILD A SECURE, COMPLIANT, AND RESILIENT FOUNDATION
BY IMPLEMENTING STRONG IDENTITY CONTROLS, PROTECTING
DATA, AND MAINTAINING CONTINUOUS VISIBILITY**

1

IAM



- Manage users, groups, and roles
- Use IAM roles for workloads
- Federate identities with AWS IAM Identity Center
- Avoid root user access
- Enable MFA for users

2

LEAST
PRIVILEGE

- Grant only necessary permissions
- Use managed policies where possible
- Create custom policies for fine-grained access
- Regularly review and remove unused access
- Follow the principle of least privilege

3

ENCRYPTION



- Encrypt data in transit using TLS
- Encrypt data at rest by default
- Use encryption for EBS, S3, RDS, and EFS
- Enable automatic encryption where available

4

KMS



- Use AWS KMS for key management
- Create customer managed keys (CMKs)
- Control key usage with key policies
- Enable key rotation
- Audit key usage regularly

5

SECRETS



- Store secrets in AWS Secrets Manager or SSM Parameter Store
- Never hardcode secrets
- Rotate secrets regularly
- Use IAM roles for application access
- Enable secret versioning

6

LOGGING AND
AUDITING

- Enable AWS CloudTrail for API activity
- Centralize logs in Amazon S3
- Use Amazon CloudWatch for monitoring
- Enable VPC Flow Logs
- Review logs regularly for anomalies

7

SECURITY
CONTROLS

- Use Security Groups and NACLs
- Enable AWS Config rules
- Use AWS WAF for web protection
- Enable GuardDuty for threat detection
- Apply patching and vulnerability management

8

COMPLIANCE
CONSIDERATIONS

- Understand your compliance requirements
- Use AWS Artifact for compliance reports
- Map workloads to compliant services
- Implement data residency controls
- Continuously monitor compliance posture

SECURITY BEST PRACTICES



**SECURE BY
DESIGN**



**PRINCIPLE OF
LEAST PRIVILEGE**



**ENCRYPT
EVERYTHING**



**MONITOR
CONTINUOUSLY**



**AUDIT
REGULARLY**



**DEFEND
IN DEPTH**

10. MIGRATION
WAVE PLANNING

PLAN AND EXECUTE MIGRATION IN CONTROLLED WAVES

TO REDUCE RISK, ENSURE DEPENDENCIES, AND DELIVER BUSINESS VALUE

1 APPLICATION
GROUPING

- Group applications by business capability
- Identify logical boundaries
- Consider technical and business relationships
- Group for efficient migration

2 DEPENDENCY-
AWARE WAVES

- Map application dependencies
- Identify upstream and downstream relationships
- Ensure correct migration order
- Avoid breaking dependencies

3 MIGRATION
PRIORITIES

- Assess business criticality
- Evaluate complexity
- Consider risk and effort
- Balance quick wins and high value
- Prioritize for maximum impact

4 PILOT
WAVE

- Start with a small, low-risk wave
- Validate migration approach
- Test tools, processes, and runbooks
- Gather lessons learned
- Refine before scale

5 WAVE
SCHEDULING

- Define wave schedule and timeline
- Align with business windows
- Plan resources and capacity
- Track progress and dependencies
- Adjust schedule as needed

6 MIGRATION
FACTORY

- Standardize tools and processes
- Reuse automation and patterns
- Build skilled, cross-functional team
- Increase speed and quality
- Ensure consistency across waves

7 MIGRATION
RUNBOOKS

- Create standard runbooks
- Include pre-checks, steps, and validation
- Define rollback procedures
- Document roles and responsibilities
- Review and improve continuously

WAVE PLANNING FLOW



KEY BENEFITS



REDUCE RISK

IMPROVE
SUCCESS RATESPEED UP
DELIVERYOPTIMIZE
RESOURCESDELIVER BUSINESS
VALUE FASTER

11. REHOST MIGRATION

WITH AWS APPLICATION MIGRATION SERVICE

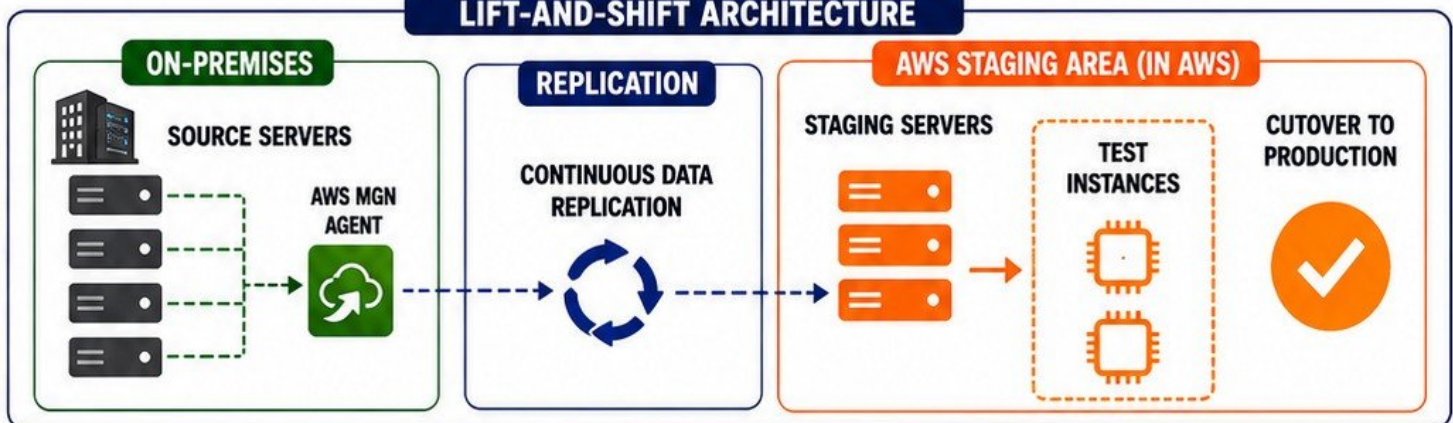


REHOST (LIFT-AND-SHIFT) MOVES YOUR APPLICATIONS TO AWS QUICKLY WITH MINIMAL CHANGES TO ARCHITECTURE.

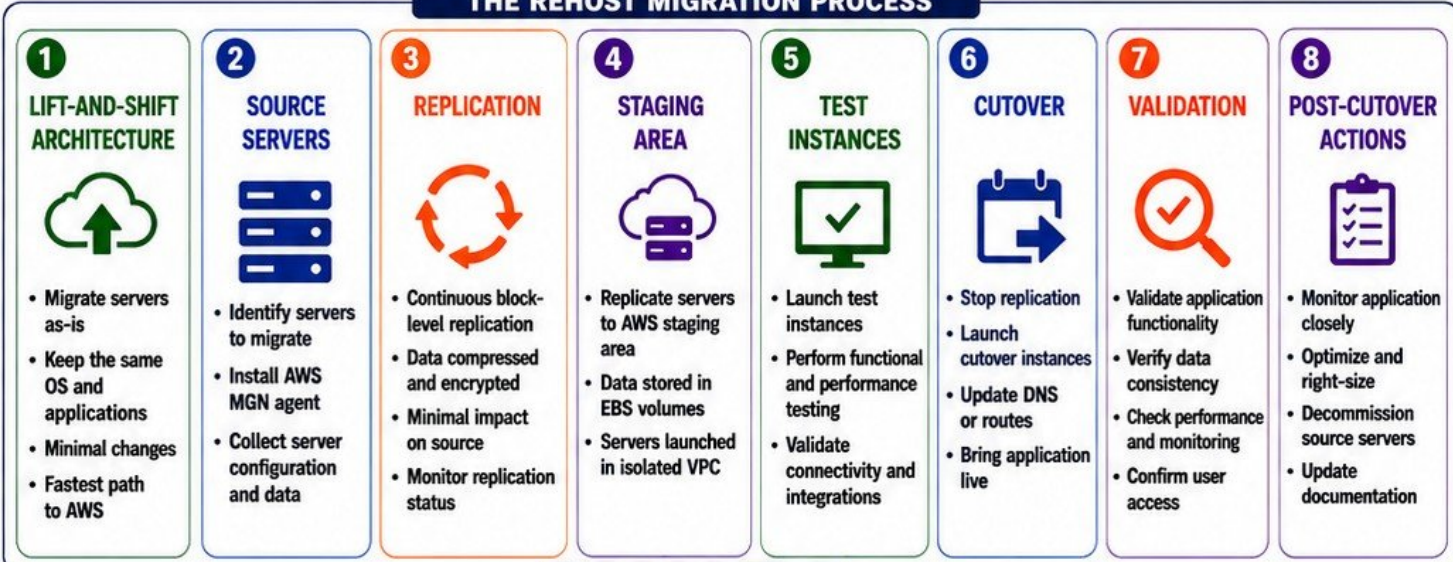


AWS
APPLICATION
MIGRATION
SERVICE

LIFT-AND-SHIFT ARCHITECTURE



THE REHOST MIGRATION PROCESS



KEY BENEFITS



FASTEST WAY
TO AWS



MINIMAL RISK
AND DISRUPTION



REDUCE MIGRATION
TIME



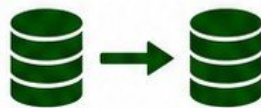
COST-EFFECTIVE
APPROACH



EASY TO SCALE
AFTER MIGRATION

12. DATABASE MIGRATION
WITH AWS DMSAWS DATABASE MIGRATION SERVICE (DMS) MAKES IT EASY TO MIGRATE
YOUR DATABASES TO AWS WITH MINIMAL DOWNTIME.AWS
DATABASE
MIGRATION
SERVICE1 DATABASE
ASSESSMENT

- Inventory all databases
- Identify database engines
- Assess database size
- Assess schema complexity
- Identify dependencies
- Evaluate migration strategy
- Estimate migration effort

2 HOMOGENEOUS
MIGRATION

- Same database engine (e.g., MySQL to MySQL)
- Simplest and fastest path
- Minimal schema changes
- Lower risk and effort
- Fully supported by AWS DMS

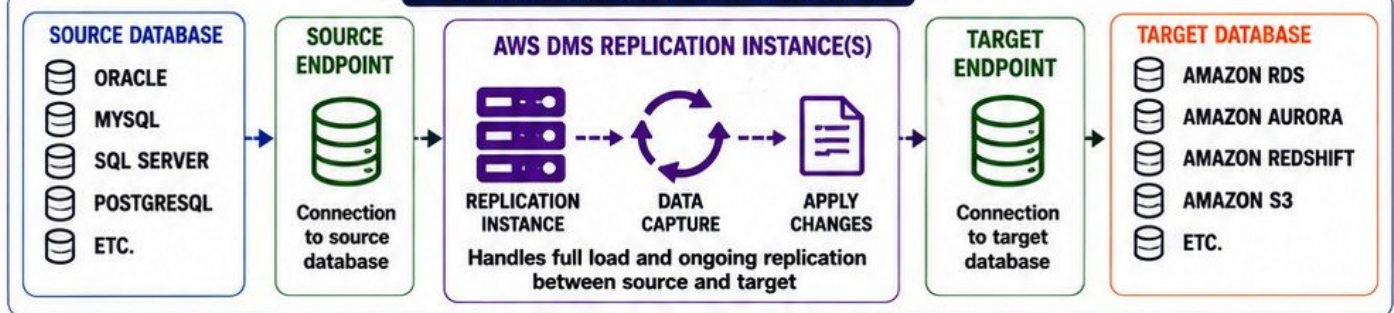
3 HETEROGENEOUS
MIGRATION

- Different database engines (e.g., Oracle to PostgreSQL)
- Use AWS DMS for data conversion
- May require schema conversion
- More testing and validation
- Higher complexity

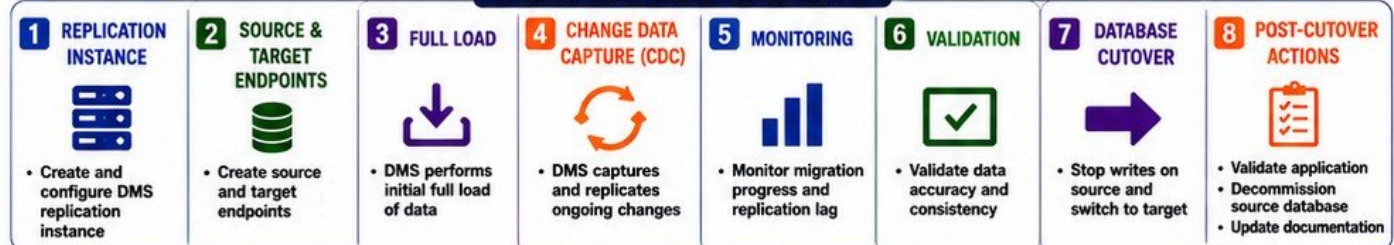
4 AWS DMS
ARCHITECTURE

- Fully managed service
- Secure and scalable
- Supports full load and ongoing replication
- Works in VPC for security
- Minimal impact on source

AWS DMS ARCHITECTURE OVERVIEW



DMS MIGRATION WORKFLOW



KEY BENEFITS



13. DATA AND STORAGE MIGRATION

MOVE YOUR DATA TO AWS SECURELY, EFFICIENTLY, AND AT SCALE

1 DATA MIGRATION PLANNING



- Define data scope
- Assess data volume
- Identify dependencies
- Plan migration waves
- Estimate time and cost
- Define success criteria

2 AMAZON S3



- Highly durable storage
- Ideal for data lakes
- Scalable and secure
- Lifecycle management
- Versioning and encryption
- Cost-effective for any size

3 AWS DATASYNC



- Fast, automated transfers
- Uses optimized protocol
- Preserves metadata
- Verifies data integrity
- Works over VPN or Direct Connect
- Ideal for scheduled syncs

4 AWS STORAGE GATEWAY



- Hybrid storage solution
- File, Volume, and Tape Gateways
- Local cache for fast access
- Seamless AWS integration
- Ideal for hybrid workloads

5 LARGE DATASET MIGRATION



- Use AWS Snow Family (Snowcone, Snowball Edge, Snowmobile)
- Ship data physically
- Ideal for petabytes of data
- Fast and cost-effective

6 SYNCHRONIZATION



- Keep data in sync
- Schedule automated jobs
- Incremental transfers
- Bi-directional sync options
- Conflict detection
- Real-time updates

7 INTEGRITY VALIDATION



- Verify data accuracy
- Checksum validation
- End-to-end verification
- AWS DataSync integrity check
- Validation reports

8 CHOOSING THE RIGHT TRANSFER APPROACH



- Consider data size
- Network bandwidth
- Urgency and downtime tolerance
- Cost and complexity
- Security requirements

DATA MIGRATION WORKFLOW

1 PLAN



Assess, plan, and define strategy

2 PREPARE



Set up AWS services and environment

3 TRANSFER



Use S3, DataSync, Storage Gateway, or Snow Family

4 SYNCHRONIZE



Keep data updated during migration window

5 VALIDATE



Verify data integrity and completeness

6 CUTOVER



Switch to AWS and confirm availability

7 OPTIMIZE



Monitor, optimize costs, and improve

CHOOSING THE RIGHT TRANSFER APPROACH

APPROACH	BEST FOR	DATA SIZE	SPEED	COST	NOTES
AMAZON S3	Cloud-native apps, data lakes, backups	Any size	Fast (Internet)	Low	Use multipart upload for large files
AWS DATASYNC	Automated transfers over network	GBs to TBs	Very Fast	Low	Uses optimized transfer protocol
AWS STORAGE GATEWAY	Hybrid workloads, local cache	Any size	Depends on network	Medium	Great for recurring access
AWS SNOW FAMILY	Very large datasets, limited bandwidth	TBs to PBs	Fast (Physical Transport)	Medium to High	Ships data securely to AWS

KEY BEST PRACTICES



ENCRYPT DATA
IN TRANSIT
AND AT REST



VALIDATE DATA
BEFORE AND
AFTER MIGRATION



USE PILOT
MIGRATIONS
FIRST



MONITOR COSTS
DURING AND AFTER
MIGRATION

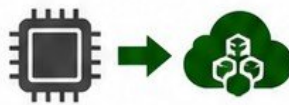


MONITOR AND
AUTOMATE
TRANSFERS

14. REPLATFORMING APPLICATIONS

IMPROVE EFFICIENCY, REDUCE COSTS, AND INCREASE AGILITY BY MOVING TO
AWS MANAGED SERVICES AND MODERN PLATFORMS.1 REHOST
VS REPLATFORM

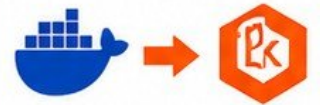
- Rehost: Lift-and-shift with minimal changes
- Replatform: Optimize using managed services and cloud features
- Replatform = Better performance, scalability, and lower cost

2 EC2 TO
MANAGED SERVICES

- Move from EC2 to AWS managed services
- Examples: ECS, EKS, Lambda, Elastic Beanstalk
- Reduces infrastructure management
- Improves scalability and availability

3 DATABASE TO
AMAZON RDS

- Migrate databases from self-managed to Amazon RDS
- Supports major engines: MySQL, PostgreSQL, Oracle, SQL Server, MariaDB, Aurora
- Automated backups, patching, and failover

4 CONTAINER
MIGRATION

- Containerize applications using Docker
- Migrate to Amazon ECS or Amazon EKS
- Improves portability, scalability, and speed of deployment

5 MANAGED
SERVICES

- Use purpose-built AWS managed services
- Reduce operational overhead
- Increase reliability and security
- Pay for what you use

6 OPERATIONAL
IMPROVEMENTS

- Less time on undifferentiated infrastructure
- Faster deployments and releases
- Automatic scaling and high availability
- Focus on innovation and business value
- Lower total cost of ownership (TCO)

7 REPLATFORMING
DECISION CRITERIA

- Business impact and goals
- Application complexity
- Dependencies and integration
- Performance requirements
- Security and compliance
- Cost and licensing
- Team skills and tooling
- Time to value

8 WHEN TO
REPLATFORM

- When managed services meet most requirements
- When operational overhead is high
- When scalability and reliability are key
- When you want to reduce costs and accelerate innovation

REPLATFORMING JOURNEY

1 ASSESS



Evaluate current application and dependencies

2 TARGET DESIGN



Design target architecture using managed services

3 MIGRATE



Migrate and adapt application to target environment

4 OPTIMIZE



Tune performance, cost, and configurations

5 OPERATE



Monitor, automate, and continuously improve

KEY BENEFITS



LOWER COSTS
Pay less and save more



BETTER PERFORMANCE
Faster, scalable applications



HIGHER RELIABILITY
Built-in HA, backups, and failover



FASTER INNOVATION
Focus on features, not infrastructure



OPERATIONAL EFFICIENCY
Automated and managed for you

15. REFACTORING AND APPLICATION MODERNIZATION

MOVE BEYOND MIGRATION TO BUILD MODERN, CLOUD-NATIVE, RESILIENT, AND SCALABLE APPLICATIONS ON AWS.

1 WHEN REFACTORING MAKES SENSE



- You need better scalability
- High operational overhead
- Frequent releases are hard
- Application is hard to change
- You want to reduce costs
- You want to leverage cloud-native services
- Business requires agility and faster innovation

2 MONOLITH MODERNIZATION



- Break down the monolith incrementally
- Identify bounded contexts
- Extract services gradually
- Use the strangler fig pattern
- Keep business value at the center
- Maintain and modernize iteratively

3 CONTAINERS



- Package applications with Docker
- Run on Amazon ECS or Amazon EKS
- Consistent environments
- Improve portability
- Faster deployments
- Better resource utilization
- Easy scaling and management

4 SERVERLESS



- Use AWS Lambda for event-driven code
- No server management
- Automatic scaling
- Pay only for what you use
- Increase agility
- Focus on business logic

5 MANAGED DATABASES



- Migrate to Amazon RDS, Aurora, or DynamoDB
- Reduce administration and maintenance
- Built-in backups and recovery
- High availability
- Better performance
- Strong security
- Automatic patching

6 DECOUPLING APPLICATIONS



- Break tightly coupled components
- Use APIs and asynchronous messaging
- Adopt event-driven architecture
- Improve resilience
- Independent scaling
- Technology flexibility
- Easier testing and deployment

7 MIGRATION VS MODERNIZATION

MIGRATION

- ✓ Lift-and-shift
- ✓ Minimal changes
- ✓ Faster to move
- ✓ Lower initial cost
- ✓ Retain existing architecture
- ✓ Short-term solution
- ✓ Focus on moving to the cloud

VS

MODERNIZATION

- ✓ Improve architecture
- ✓ Use cloud-native services
- ✓ More time and effort
- ✓ Higher long-term value
- ✓ Better scalability, security, and agility
- ✓ Future-ready
- ✓ Transform the way you build and operate

MODERNIZATION PATH



ASSESS

Assess current state and pain points



PLAN

Define modernization strategy and roadmap



EXECUTE

Refactor, adopt services, and modernize iteratively



VALIDATE

Test, validate, and ensure reliability



OPERATE

Monitor, optimize, and continuously improve

KEY BENEFITS



INCREASE AGILITY

Respond to business needs faster



REDUCE COSTS

Optimize resources and pay less



IMPROVE RELIABILITY

Build resilient and highly available apps



BETTER PERFORMANCE

Deliver faster and more efficiently



DRIVE INNOVATION

Focus on creating business value

16. TESTING BEFORE MIGRATION CUTOVER



**VALIDATE THE ENVIRONMENT, APPLICATIONS, AND DATA
TO ENSURE A SUCCESSFUL AND RISK-FREE CUTOVER**

1 FUNCTIONAL TESTING



- Verify all features and functions work as expected
- Validate business workflows
- Check user roles and permissions
- Ensure data is displayed correctly

2 INTEGRATION TESTING



- Validate interactions between components
- Test APIs and service integrations
- Verify data flow across systems
- Ensure third-party integrations work

3 PERFORMANCE TESTING



- Test under expected and peak load
- Validate response times and throughput
- Identify performance bottlenecks
- Ensure scalability requirements are met

4 SECURITY VALIDATION



- Verify IAM roles and least privilege access
- Test encryption in transit and at rest
- Scan for vulnerabilities
- Validate security controls and policies

5 NETWORK VALIDATION



- Verify connectivity between all tiers
- Test DNS resolution and routing
- Validate VPN / Direct Connect (if applicable)
- Check firewall and security group rules

6 DATABASE VALIDATION



- Validate data integrity and accuracy
- Test database performance
- Verify backups and recovery
- Check relationships and constraints

7 USER ACCEPTANCE TESTING



- Validate with real business users
- Ensure requirements are fully met
- Collect feedback and resolve issues
- Obtain final user approval

8 GO / NO-GO DECISION



- Review all test results
- Assess risks and open issues
- Confirm readiness with stakeholders
- Approve cutover or delay the migration

TESTING BEST PRACTICES



CREATE A
TEST PLAN



INVOLVE USERS
EARLY



AUTOMATE
WHERE POSSIBLE



DOCUMENT AND
TRACK ISSUES



RETEST AFTER
FIXES



DO NOT SKIP
VALIDATIONS



**THOROUGH TESTING MINIMIZES RISK, ENSURES BUSINESS CONTINUITY,
AND BUILDS CONFIDENCE FOR A SUCCESSFUL CUTOVER.**

17. PRODUCTION
CUTOVEREXECUTE THE CUTOVER WITH PRECISION.
MINIMIZE RISK. ENSURE CONTINUITY. DELIVER VALUE.

1

CUTOVER
RUNBOOK

- Follow the approved cutover runbook step-by-step
- Assign clear roles and responsibilities
- Have rollback plan and contacts ready

2

CHANGE
FREEZE

- Implement a change freeze before cutover
- No new changes, deployments, or configuration updates
- Communicate freeze window to all teams

3

FINAL
SYNCHRONIZATION

- Complete final data synchronization
- Ensure databases, files, and queues are in sync
- Verify no pending transactions or replication lag

4

DNS
CHANGES

- Update DNS records to point to AWS environment
- Reduce TTL before cutover to speed propagation
- Verify DNS resolution from multiple locations

5

TRAFFIC
MIGRATION

- Route user traffic to the AWS environment
- Monitor traffic flow and system metrics
- Watch for errors, latency, and anomalies

6

APPLICATION
STARTUP

- Start all application services in the correct order
- Verify dependencies and configurations
- Confirm healthy status of all components

7



VALIDATION

- Validate application functionality end-to-end
- Verify data accuracy and business transactions
- Confirm performance and key metrics are normal

8

STAKEHOLDER
COMMUNICATION

- Communicate cutover start and completion
- Provide status updates during and after cutover
- Confirm success and share next steps

A SUCCESSFUL CUTOVER IS THE RESULT OF PREPARATION,
PRECISION, AND CLEAR COMMUNICATION.
PLAN IT. TEST IT. EXECUTE IT. OWN IT.

18. ROLLBACK AND MIGRATION FAILURE RECOVERY



**BE PREPARED TO RECOVER QUICKLY.
MINIMIZE DOWNTIME, DATA LOSS, AND BUSINESS IMPACT.**

1 ROLLBACK CRITERIA



- Define clear rollback criteria before cutover
- Set measurable thresholds (SLA, errors, performance, data)
- Include business impact and risk levels
- Get stakeholder approval

2 FAILED CUTOVER



- Detect failure quickly using monitoring and alerts
- Stop traffic to new environment
- Communicate failure immediately
- Trigger rollback plan execution

3 DATA CONSISTENCY



- Ensure last successful data sync point
- Validate data integrity before rollback
- Handle in-flight transactions
- Prevent data loss or corruption

4 DNS ROLLBACK



- Revert DNS to previous environment
- Use low TTL values before cutover
- Verify DNS propagation globally
- Confirm users reach the old environment

5 APPLICATION ROLLBACK



- Redeploy previous stable application version
- Restore configuration and environment
- Verify dependencies and integrations
- Confirm application health

6 DATABASE ROLLBACK CONSIDERATIONS



- Decide between point-in-time recovery or full restore
- Consider replication lag and backups
- Roll back schema changes if needed
- Validate database consistency after restore

7 INCIDENT ESCALATION



- Escalate to incident response team
- Follow escalation matrix and contact list
- Keep stakeholders informed
- Document all actions and decisions

8 POSTMORTEM



- Conduct postmortem after recovery
- Identify root cause
- Document lessons learned
- Update runbooks and processes
- Share improvements with the team



**A STRONG ROLLBACK PLAN IS YOUR SAFETY NET.
PLAN AHEAD. ACT FAST. RECOVER SMART.**

ROLLBACK BEST PRACTICES



PLAN
AHEAD



TEST
ROLLBACK
PLAN



MONITOR
CONTINUOUSLY



COMMUNICATE
CLEARLY



IMPROVE
CONTINUOUSLY

19. POST-MIGRATION OPERATIONS AND OPTIMIZATION



CONTINUOUSLY OPERATE, OPTIMIZE, AND IMPROVE
TO DELIVER RELIABILITY, PERFORMANCE, AND BUSINESS VALUE.

1

MONITORING



- Monitor system health and availability
- Track KPIs, SLIs, and SLOs
- Create dashboards and custom metrics
- Set alarms and notifications
- Proactively detect and resolve issues

2

CLOUDWATCH



- Use Amazon CloudWatch for metrics and alarms
- Monitor logs, events, and resources
- Create dashboards and insights
- Automate alerting and actions

3

LOGGING



- Centralize logs with CloudWatch Logs
- Enable log retention and encryption
- Create log insights and queries
- Integrate with monitoring and alerting

4

BACKUP



- Implement automated backups
- Test backup restoration regularly
- Store backups securely
- Define retention policies
- Monitor backup success

5

DISASTER RECOVERY



- Define RPO and RTO targets
- Implement multi-AZ / multi-region strategies
- Test DR plans regularly
- Document and maintain DR runbooks
- Continuously improve DR posture

6

PERFORMANCE TUNING



- Monitor performance metrics
- Identify and remove bottlenecks
- Optimize queries and code
- Use caching and CDN
- Load test and validate improvements

7

COST OPTIMIZATION



- Monitor and analyze costs
- Eliminate wasted resources
- Use budgets and alerts
- Leverage savings plans and reserved instances
- Review costs regularly

8

RIGHTSIZING



- Rightsize compute, storage, and databases
- Use performance data to make decisions
- Adjust capacity based on demand
- Review and optimize continuously

9

OPERATIONAL HANDOVER



- Transfer knowledge and documentation
- Establish operational runbooks
- Define roles and responsibilities
- Train operations and support teams
- Confirm smooth transition and support

POST-MIGRATION SUCCESS MEANS CONTINUOUS IMPROVEMENT



RELIABILITY
YOU CAN TRUST



PERFORMANCE
THAT SCALES



SECURITY
BY DESIGN



COSTS
UNDER CONTROL



OPERATIONS
THAT DELIVER VALUE

20. ENTERPRISE MIGRATION LAB AND FINAL CHECKLIST



A HANDS-ON LAB TO BRING EVERYTHING TOGETHER.
PLAN, SIMULATE, EXECUTE, AND OPTIMIZE YOUR MIGRATION.

1 ASSESS AN EXAMPLE ON-PREMISES ENVIRONMENT



- Review environment details
- Identify applications, databases, and services
- Capture infrastructure and business requirements
- Document pain points and goals

2 CHOOSE A 7 R STRATEGY



- Evaluate all 7 R options
- Select the best strategy per application
- Justify choices with business and technical rationale
- Document strategy decisions

3 MAP DEPENDENCIES



- Identify application dependencies
- Map network, data, and service dependencies
- Create dependency diagrams
- Identify critical paths and risks

4 DESIGN TARGET AWS ARCHITECTURE



- Design architecture on AWS
- Select appropriate AWS services
- Address security, HA, scalability, and resilience
- Create architecture diagrams

5 CREATE MIGRATION WAVES



- Group applications into migration waves
- Define wave goals and priorities
- Identify dependencies between waves
- Create wave timeline and criteria

6 BUILD MIGRATION RUNBOOK



- Create step-by-step migration runbook
- Include pre-migration tasks
- Include cutover steps
- Include rollback steps
- Define roles and responsibilities

7 SIMULATE MIGRATION



- Perform dry run of migration
- Test data transfer and application setup
- Validate runbook steps
- Document issues and improvements

8 EXECUTE CUTOVER



- Execute cutover per runbook
- Monitor systems closely
- Verify services are running as expected
- Communicate status to stakeholders

9 VALIDATE PRODUCTION



- Validate application functionality
- Verify data integrity
- Check performance and availability
- Confirm security controls
- Obtain business sign-off

10 SIMULATE ROLLBACK



- Trigger rollback simulation
- Follow rollback runbook
- Validate systems return to previous state
- Document results and gaps

11 POST-MIGRATION CHECKLIST



- Review monitoring and alerts
- Validate backups and DR
- Optimize performance and costs
- Update documentation
- Close migration tasks

12 SENIOR ENGINEER RECOMMENDATIONS



- Lessons learned
- Best practices
- Common pitfalls to avoid
- Architecture and operational advice
- Continuous improvement recommendations



COMPLETE THE LAB. FOLLOW THE CHECKLIST.
DELIVER A SUCCESSFUL MIGRATION.



CLEAR PLAN
AND STRATEGY



SKILLED TEAM
AND ROLES



TESTING AND
VALIDATION



MONITORING
AND OPTIMIZATION



COMMUNICATION
AND ALIGNMENT